LOCKWOOD STEM

Aerospace Academy Version

Unit 6

Assemblies, Mechanisms & Aerospace Systems

Unit Overview

Students move from single parts to multi-part systems by designing assemblies and mechanisms. They learn joints, exploded views, BOMs, assembly instructions, motion, clearances, and system-level design thinking.

Essential Question

How do engineers design parts that work together as a complete aerospace system?

Name: ______________________ **Period:** ______________________ **Date:** ______________________

Learning Targets

1. Explain the major concepts of assemblies, mechanisms & aerospace systems.
2. Apply engineering documentation, sketching, modeling, testing, or communication skills to an aerospace-focused task.
3. Use evidence and reflection to improve design work.
4. Submit a complete unit portfolio or project package.

Unit Deliverables

- Mechanism research
- Assembly concept sketches
- CAD assembly or paper prototype
- Joint/clearance notes
- Exploded view
- BOM
- Assembly instructions
- System reflection

Vocabulary

Term	Definition
Assembly	
Component	
Joint	
Constraint	
Mechanism	
Degree of Freedom	
Clearance	
Interference	
Exploded View	
Assembly Instructions	
BOM	
Subsystem	
Motion Study	
Interface	
Maintenance	

Unit Project: Aerospace Mechanism Assembly

You will apply the skills from this unit to create an aerospace-focused project package. Your work should show clear documentation, professional effort, evidence-based decisions, and reflection.

Day 1 — From Parts to Systems

Objective: I can explain how individual parts work together in systems.
Bellwork: Why is a good part not enough by itself?

Core Task: Analyze a simple aerospace assembly and identify components and interfaces.

Evidence of Work / Notes:

Exit Ticket: What is an interface?

Day 2 — Mechanisms in Aerospace

Objective: I can identify mechanisms used in aerospace systems.
Bellwork: Where do aircraft or spacecraft need moving parts?

Core Task: Research control surfaces, landing gear, payload doors, gimbals, or deployment systems.

Evidence of Work / Notes:

Exit Ticket: What mechanism interests you?

Day 3 — Degrees of Freedom

Objective: I can describe how parts move or are constrained.
Bellwork: How many ways can an object move?

Core Task: Use examples to identify rotation, translation, fixed joints, and limited motion.

Evidence of Work / Notes:

Exit Ticket: What motion does your mechanism need?

Day 4 — Clearance and Interference

Objective: I can explain why moving parts need clearance.
Bellwork: What happens if parts overlap or rub?

Core Task: Analyze examples of clearance, interference, and tolerance issues.

Evidence of Work / Notes:

Exit Ticket: Where will your design need clearance?

Day 5 — Assembly Concept Sketches

Objective: I can sketch a multi-part aerospace mechanism.
Bellwork: How do sketches show how parts connect?

Core Task: Create at least three assembly concepts with labeled components and motion arrows.

Evidence of Work / Notes:

Exit Ticket: What component controls motion?

Day 6 — Selecting a Mechanism

Objective: I can select an assembly concept using criteria.
Bellwork: What makes one mechanism better than another?

Core Task: Compare concepts using criteria such as reliability, simplicity, size, and manufacturability.

Evidence of Work / Notes:

Exit Ticket: Which concept did you select?

Day 7 — Component Planning

Objective: I can plan the parts needed for an assembly.
Bellwork: Why should parts be listed before modeling?

Core Task: Create a component list and describe each part's function.

Evidence of Work / Notes:

Exit Ticket: What part is most critical?

Day 8 — CAD Assembly Setup

Objective: I can organize CAD components for an assembly.
Bellwork: Why should components be separate in CAD?

Core Task: Create or plan CAD components and organize files.

Evidence of Work / Notes:

Exit Ticket: How will you name your components?

Day 9 — Joints and Motion

Objective: I can apply or describe joints that control motion.
Bellwork: What joint matches your mechanism?

Core Task: Add joints or diagram motion for the mechanism.

Evidence of Work / Notes:

Exit Ticket: What motion did you create?

Day 10 — Assembly Work Day 1

Objective: I can build or model the main assembly.
Bellwork: What must line up for the system to work?

Core Task: Build/model base, supports, pivots, and moving components.

Evidence of Work / Notes:

Exit Ticket: What did you complete?

Day 11 — Assembly Work Day 2

Objective: I can refine fit and movement in an assembly.
Bellwork: What prevents smooth motion?

Core Task: Revise clearances, pivots, interfaces, and mounting features.

Evidence of Work / Notes:

Exit Ticket: What issue did you fix?

Day 12 — Exploded Views

Objective: I can create an exploded view to show assembly order.
Bellwork: Why are exploded views useful?

Core Task: Create an exploded sketch, CAD view, or diagram.

Evidence of Work / Notes:

Exit Ticket: What part is installed first?

Day 13 — Assembly Instructions

Objective: I can write clear assembly instructions.
Bellwork: What makes directions easy to follow?

Core Task: Write step-by-step instructions with parts, tools, and checks.

Evidence of Work / Notes:

Exit Ticket: What step needs a warning or note?

Day 14 — System Review

Objective: I can evaluate an assembly as a complete system.
Bellwork: What can fail in a multi-part system?

Core Task: Check function, motion, fit, maintainability, and documentation.

Evidence of Work / Notes:

Exit Ticket: What system risk did you identify?

Day 15 — Assembly Submission

Objective: I can submit and reflect on a complete assembly design.
Bellwork: How is designing an assembly different from designing a part?

Core Task: Submit assembly documentation and complete reflection.

Evidence of Work / Notes:

Exit Ticket: What did you learn about mechanisms?

Unit 6 Submission Checklist

Item	Complete?
Mechanism research	❏
Assembly concept sketches	❏
CAD assembly or paper prototype	❏
Joint/clearance notes	❏
Exploded view	❏
BOM	❏
Assembly instructions	❏
System reflection	❏

Unit 6 Project Rubric

Category	Points
Technical accuracy and completeness	20
Engineering process and documentation	20
Aerospace connection and design intent	15
Quality of sketches, models, data, or drawings	20
Use of evidence, testing, or decision making	15
Reflection and professionalism	10
Total	**100**

Unit 6 Student Work Templates

Assembly Concept Sketch

Purpose: Isometric sketching
Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name:	____________________	**Scale:**	____________________
Purpose:	____________________	**Date:**	____________________

Notes / annotations: ____

Mechanism Motion Diagram

Purpose: Isometric sketching

Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name:	______________	**Scale:**	______________
Purpose:	______________	**Date:**	______________

Notes / annotations: ______________________________

Exploded View Planning Page

Purpose: Isometric sketching

Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name: ____________________ **Scale:** ____________________

Purpose: ____________________ **Date:** ____________________

Notes / annotations: ____________________

Bill of Materials and Revision Template

Purpose: Documentation
Directions: List parts, materials, quantities, manufacturing methods, and revision notes so someone else could understand what changed.

Item	Part Name	Material / Process	Qty.	Notes
1				
2				
3				
4				
5				
6				

Rev.	Date	Change Made	Reason for Change
A			
B			
C			

Assembly Procedure Template

Purpose: Technical communication
Directions: Write clear steps so another student could assemble the system safely and correctly.

Step	Action	Check / Warning
1		
2		
3		
4		
5		
6		
7		
8		

Assembly notes: